Online on xv6 System Call

Section: A2 — **Time:** 30 minutes

In this assignment, you will maintain a per-process LIFO/ring buffer inside `struct proc` that records details about the process's last 3 executed system calls. A user-space program will fetch this history struct using kernel space memory copying.

Required System Call:

1. `getAuditTrail(struct audit_trail *at)` : Copies the calling process's recent system call history into the provided user space struct pointer. Returns 0 on success and -1 on invalid memory address.

The structure definitions are as follows:

```
struct syscall_record {
    int syscall_num;     // System call number (from syscall.h)
    int return_val;      // Return value of the executed syscall
};

struct audit_trail {
    int count;                          // Total records stored (max 3)
    struct syscall_record records[3]; // Most recent calls (index 0 = newest)
};
```

Mechanics:

- Update `syscall.c` and `struct proc` so that every time a system call returns for a process, its syscall number and return value are logged into its internal 3-element buffer.
- The system call `getAuditTrail` itself should **not** log its own execution into the buffer to avoid mutating the history while reading it.

Required User Command:

2. `audit_show`

See the sample I/O for clarification:

```
$ audit_show
-- Recent Syscalls --
1. Syscall #15 (open) -> Ret: 3
2. Syscall #5 (read) -> Ret: 1024
3. Syscall #21 (close) -> Ret: 0
```

Submission:

```
git add --all
git diff HEAD > ../{studentID}.patch
```